

CDKN2B fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes CDKN2B gene fusions from the msk_impact_50k_2026 study, covering 12 fusion events. 2/3 fusions with known frame status (66.7%) were in-frame. 2/10 fusions (20.0%) retained the Ankyrin repeats (3 copies) domain. Domain retention was tested with Fisher's exact test ($p=0.0222222$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for CDKN2B.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 12 analyzed fusion events, identifying 7 distinct fusion partner genes. The most frequent partner was CDKN2A, observed in 6 events.

Domain retention

Ankyrin repeats (3 copies) domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 2/2 (100.0%) of in-frame fusions retained the domain, $p=0.0222222$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.041958.

Domain disruption

Domain-disruption analysis was skipped: CDKN2B has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 10 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 53 aa (exon 2; permutation-corrected $p=0.762376$, not statistically significant at $\alpha=0.05$). This is -1 aa from the nearest configured domain boundary at 54 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=2$) and "not_retained" ($n=8$). For Frame_status, retained: 2/2 (100.0%, 95% CI 34.2%-100.0%); not_retained: 0/1 (0.0%, 95% CI 0.0%-79.3%). Welch's t-test comparing Tumor_variant_count between groups gave means of 11.5 vs 48.25, $p=0.170234$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for CDKN2B in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (3 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 2 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: CDKN2B has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: CDKN2B has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (20 row(s) in window_scan, 8 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CDKN2A	6	0.5	0.13771847870814183	None	0.6721311475409837	0.2744647709419452	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.39086971112797186	1
CNTRL	1	0.08333333333333333	0.13771847870814183	None	1.0	0.2744647709419452	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.2960951209640374	2
MTAP	1	0.08333333333333333	0.13771847870814183	None	1.0	0.2744647709419452	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.2960951209640374	3
SLC24A2	1	0.08333333333333333	0.13771847870814183	None	0.7540983606557377	0.2744647709419452	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.24998856358698826	4
GAB2	1	0.08333333333333333	0.13771847870814183	None	0.6721311475409837	0.2744647709419452	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.23461971112797192	5
ELAVL2	1	0.08333333333333333	0.13771847870814183	None	None	0.2744647709419452	confidence_certainty, domain_retention, recurrence	0.13365553349419998	6
CDKN2B-AS1	1	0.08333333333333333	0.13771847870814183	None	0.0	0.2744647709419452	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.10859512096403748	7

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
38	0	2	1	7	0.0	1.0	-0.0
52	0	2	3	5	0.0	1.0	-0.0
53	2	0	3	5	None	0.44444444444444453	0.3521825181113624
69	2	0	4	4	None	0.4666666666666668	0.3309932190414243
73	2	0	5	3	None	1.0	-0.0
75	2	0	6	2	None	1.0	-0.0

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
113	2	0	7	1	None	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

2	0
0	8

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	Ankyrin repeats (3 copies)	10	2	5	3	0.34104938271604934

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
2	2	12	0.16666666666666666

frequency tables

Partner_gene_counts

Partner_gene	Event_count
CDKN2A	6
CDKN2B-AS1	1
CNTRL	1
ELAVL2	1
GAB2	1
MTAP	1
SLC24A2	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
9	22006000	22007000	5	EVT-P-0009347-T01-IM5-34, EVT-P-0010183-T02-IM6-10, EVT-P-0035576-T01-IM6-2, EVT-P-0045019-T02-IM6-12, EVT-P-0053040-T01-IM6-38
9	22008000	22009000	4	EVT-P-0002886-T01-IM3-40, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, EVT-P-0060581-T01-IM7-11
9	22007000	22008000	1	EVT-P-0038263-T01-IM6-13

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
52	2	2	489	False
53	2	2	42	False

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
-147	53	200	5	5	2	0	None	0.44444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0038263-T01-IM6-13, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, EVT-P-0060581-T01-IM7-11
-131	69	200	6	4	2	0	None	0.4666666666666668	0.3309932190414243	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-10, EVT-P-0038263-T01-IM6-13, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, ... (6 total)
-47	53	100	5	5	2	0	None	0.44444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0038263-T01-IM6-13, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, EVT-P-0060581-T01-IM7-11
-31	69	100	6	4	2	0	None	0.4666666666666668	0.3309932190414243	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-10, EVT-P-0038263-T01-IM6-13, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, ... (6 total)
3	53	50	5	5	2	0	None	0.44444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0038263-T01-IM6-13, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, EVT-P-0060581-T01-IM7-11

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
19	69	50	6	4	2	0	None	0.4666666666666668	0.3309932190414243	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-10, EVT-P-0038263-T01-IM6-13, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, ... (6 total)
28	53	25	5	5	2	0	None	0.4444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0038263-T01-IM6-13, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, EVT-P-0060581-T01-IM7-11
38	63	25	5	5	2	0	None	0.4444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0038263-T01-IM6-13, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, EVT-P-0060581-T01-IM7-11
44	69	25	5	5	2	0	None	0.4444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-10, EVT-P-0038263-T01-IM6-13, EVT-P-0042985-T01-IM6-37, EVT-P-0060581-T01-IM7-11
48	73	25	6	4	2	0	None	0.4666666666666668	0.3309932190414243	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-10, EVT-P-0038263-T01-IM6-13, EVT-P-0042985-T01-IM6-37, EVT-P-0053040-T01-IM6-38, ... (6 total)
53	78	25	5	5	2	0	None	0.4444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-10, EVT-P-0035576-T01-IM6-2, EVT-P-0042985-T01-IM6-37, EVT-P-0053040-T01-IM6-38
53	103	50	5	5	2	0	None	0.4444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-10, EVT-P-0035576-T01-IM6-2, EVT-P-0042985-T01-IM6-37, EVT-P-0053040-T01-IM6-38
63	113	50	4	6	0	2	0.0	0.4666666666666668	0.3309932190414243	EVT-P-0010183-T02-IM6-10, EVT-P-0035576-T01-IM6-2, EVT-P-0045019-T02-IM6-12, EVT-P-0053040-T01-IM6-38
64	114	50	5	5	0	2	0.0	0.4444444444444453	0.3521825181113624	EVT-P-0009347-T01-IM5-34, EVT-P-0010183-T02-IM6-10, EVT-P-0035576-T01-IM6-2, EVT-P-0045019-T02-IM6-12, EVT-P-0053040-T01-IM6-38
69	119	50	5	5	0	2	0.0	0.4444444444444453	0.3521825181113624	EVT-P-0009347-T01-IM5-34, EVT-P-0010183-T02-IM6-10, EVT-P-0035576-T01-IM6-2, EVT-P-0045019-T02-IM6-12, EVT-P-0053040-T01-IM6-38

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
69	169	100	5	5	0	2	0.0	0.4444444444444453	0.3521825181113624	EVT-P-0009347-T01-IM5-34, EVT-P-0010183-T02-IM6-10, EVT-P-0035576-T01-IM6-2, EVT-P-0045019-T02-IM6-12, EVT-P-0053040-T01-IM6-38
69	269	200	5	5	0	2	0.0	0.4444444444444453	0.3521825181113624	EVT-P-0009347-T01-IM5-34, EVT-P-0010183-T02-IM6-10, EVT-P-0035576-T01-IM6-2, EVT-P-0045019-T02-IM6-12, EVT-P-0053040-T01-IM6-38
73	123	50	4	6	0	2	0.0	0.4666666666666668	0.3309932190414243	EVT-P-0009347-T01-IM5-34, EVT-P-0035576-T01-IM6-2, EVT-P-0045019-T02-IM6-12, EVT-P-0053040-T01-IM6-38
73	173	100	4	6	0	2	0.0	0.4666666666666668	0.3309932190414243	EVT-P-0009347-T01-IM5-34, EVT-P-0035576-T01-IM6-2, EVT-P-0045019-T02-IM6-12, EVT-P-0053040-T01-IM6-38
73	273	200	4	6	0	2	0.0	0.4666666666666668	0.3309932190414243	EVT-P-0009347-T01-IM5-34, EVT-P-0035576-T01-IM6-2, EVT-P-0045019-T02-IM6-12, EVT-P-0053040-T01-IM6-38

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
28	53	25	5	5	2	0	None	0.4444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0038263-T01-IM6-13, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, EVT-P-0060581-T01-IM7-11
44	69	25	5	5	2	0	None	0.4444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-10, EVT-P-0038263-T01-IM6-13, EVT-P-0042985-T01-IM6-37, EVT-P-0060581-T01-IM7-11
53	78	25	5	5	2	0	None	0.4444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-10, EVT-P-0035576-T01-IM6-2, EVT-P-0042985-T01-IM6-37, EVT-P-0053040-T01-IM6-38
64	114	50	5	5	0	2	0.0	0.4444444444444453	0.3521825181113624	EVT-P-0009347-T01-IM5-34, EVT-P-0010183-T02-IM6-10, EVT-P-0035576-T01-IM6-2, EVT-P-0045019-T02-IM6-12, EVT-P-0053040-T01-IM6-38
48	73	25	6	4	2	0	None	0.4666666666666668	0.3309932190414243	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-10, EVT-P-0038263-T01-IM6-13, EVT-P-0042985-T01-IM6-37, EVT-P-0053040-T01-IM6-38, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
19	69	50	6	4	2	0	None	0.4666666666666668	0.3309932190414243	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-10, EVT-P-0038263-T01-IM6-13, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, ... (6 total)
63	113	50	4	6	0	2	0.0	0.4666666666666668	0.3309932190414243	EVT-P-0010183-T02-IM6-10, EVT-P-0035576-T01-IM6-2, EVT-P-0045019-T02-IM6-12, EVT-P-0053040-T01-IM6-38
73	123	50	4	6	0	2	0.0	0.4666666666666668	0.3309932190414243	EVT-P-0009347-T01-IM5-34, EVT-P-0035576-T01-IM6-2, EVT-P-0045019-T02-IM6-12, EVT-P-0053040-T01-IM6-38

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0035576-T01-IM6-2	P-0035576-T01-IM6		Gastrointestinal Stromal Tumor	GIST	CDKN2A-CDKN2B	CDKN2A	unknown	five_prime	22006179	2	75	False	74	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.2716049382716049	True			Ankyrin repeats (3 copies)	CDKN2B (NM_004936) - CDKN2A (NM_0040077) Rearrangement: c. 224:CDKN2B_c.153:CDKN2AAdel Protein Fusion: mid-exon {CDKN2B:CDKN2A}	NA
EVT-P-0010183-T02-IM6-10	P-0010183-T02-IM6		Gastrointestinal Stromal Tumor	GIST	CDKN2B-CDKN2A	CDKN2A	unknown	five_prime	22006197	2	69	False	68	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.19753086419753085	True			Ankyrin repeats (3 copies)	CDKN2B (NM_004936) - CDKN2A (NM_0058195) rearrangement: c. 206:CDKN2B_c.194-11:CDKN2Adel Protein Fusion: mid-exon {CDKN2B:CDKN2A}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0060581-T01-IM7-11	P-0060581-T01-IM7		Hepatobiliary Cancer	HCC	CDKN2B-CDKN2A	CDKN2A	out-of-frame	five_prime	22008168	1	52	True	52	True	exact match	lost	0.0	False		Ankyrin repeats (3 copies)		CDKN2B (NM_004936) - CDKN2A (NM_000077) rearrangement: c. 156+629:CDKN2B_c.458-143:CDKN2A Protein Fusion: out of frame {CDKN2B: CDKN2A}	NA
EVT-P-0045019-T02-IM6-12	P-0045019-T02-IM6		Non-Small Cell Lung Cancer	LUAD	CDKN2B-CDKN2A	CDKN2A	unknown	five_prime	22006065	2	113	False	112	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.7407407407407407	True			Ankyrin repeats (3 copies)	CDKN2B (NM_004936) - CDKN2A (NM_0058195) rearrangement: c. 338:CDKN2B_c.303:CDKN2A Protein Fusion: mid-exon {CDKN2B:CDKN2A}	NA

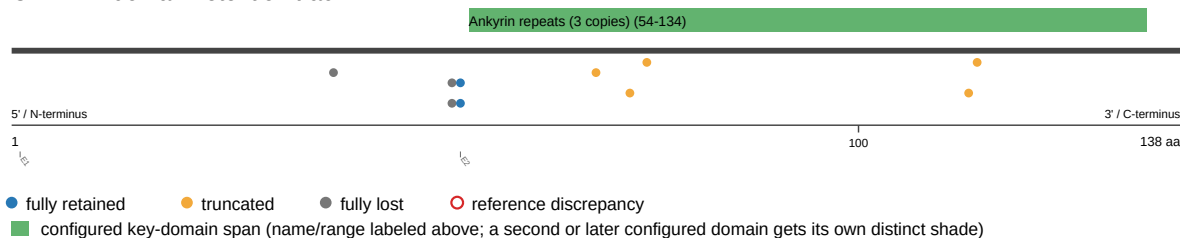
event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0038263-T01-IM6-13	P-0038263-T01-IM6		Endometrial Cancer	UMEC	CDKN2B-CDKN2A	CDKN2A	unknown	five_prime	22007679	1	52	True	52	True	exact match	lost	0.0	False		Ankyrin repeats (3 copies)		CDKN2B (NM_004936) - CDKN2A (NM_000077) Rearrangement: c. 156+1118:CDKN2B_c.192:CDKN2Adel Protein Fusion: mid-exon {CDKN2B:CDKN2A}	NA
EVT-P-0009347-T01-IM5-34	P-0009347-T01-IM5		Renal Cell Carcinoma	CCRCC	CDKN2B-CDKN2B-AS1	CDKN2B-AS1	unknown	five_prime	22006062	2	114	False	113	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.7530864197530864	True			Ankyrin repeats (3 copies)	CDKN2B (NM_004936) rearrangement: c. 341_chr9:g.22015263inv Protein fusion: mid-exon (CDKN2B-CDKN2BAS)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0045616-T01-IM6-36	P-0045616-T01-IM6		Pancreatic Cancer	PAAD	CDKN2B-SLC24A2	SLC24A2	unknown	five_prime	22008840	1	38	False	37	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Ankyrin repeats (3 copies)		CDKN2B (NM_004936) - SLC24A2 (NM_020344) rearrangement: c.113:CDKN2B_c.931-5132:SLC24A2del Protein Fusion: mid-exon {CDKN2B:SLC24A2}	NA
EVT-P-0042985-T01-IM6-37	P-0042985-T01-IM6		Prostate Cancer	PRAD	CNTRL-CDKN2B	CNTRL	in-frame	three_prime	22008250	2	53	True	53	True	exact match	retained	1.0	False	Ankyrin repeats (3 copies)			CNTRL (NM_007018) - CDKN2B (NM_004936) rearrangement: c.1455+114:CNTRL_c.156+547:CDKN2Binv Protein Fusion: in frame {CNTRL:CDKN2B}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0053040-T01-IM6-38	P-0053040-T01-IM6		Bone Cancer	OS	GAB2-CDKN2B	GAB2	unknown	three_prime	22006184	2	73	False	73	True	exact match	disrupted	0.7654320987654321	True			Ankyrin repeats (3 copies)	GAB2 (NM_080491) - CDKN2B (NM_004936) rearrangement: t(9;11)(p21.3;q14.1)(chr9:g.22006184::chr11:g.78068634) Protein Fusion: mid-exon {GAB2:CDKN2B}	NA
EVT-P-002886-T01-IM3-40	P-0002886-T01-IM3		Non-Small Cell Lung Cancer	LUAD	MTAP-CDKN2B	MTAP	in-frame	three_prime	22008292	2	53	True	53	True	exact match	retained	1.0	False	Ankyrin repeats (3 copies)			NA Protein fusion: in frame (MTAP-CDKN2B)	NA

domain retention outliers

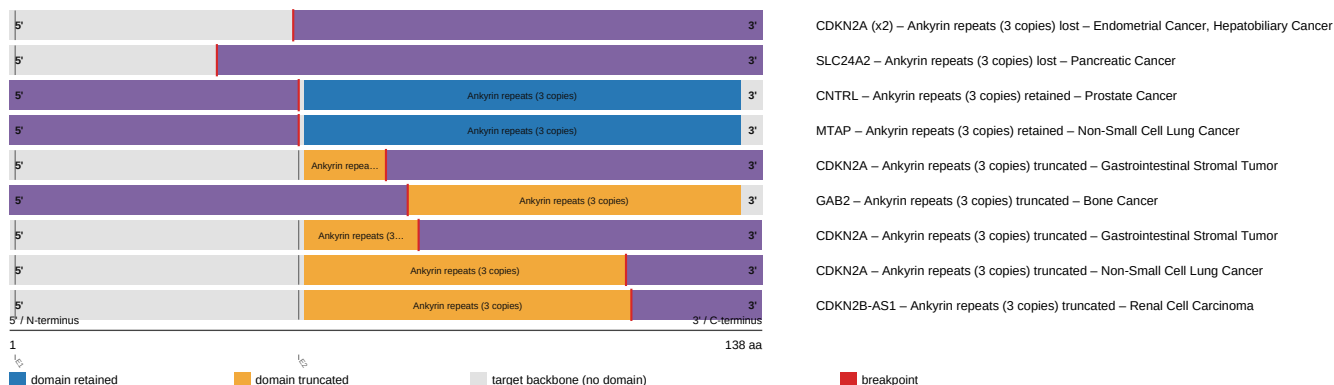
CDKN2B domain-retention track



fusion schematic

CDKN2B fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (CDKN2B portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative: row labels identify partners.